Advanced database management systems

Topic 5: Graph databases

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Overview

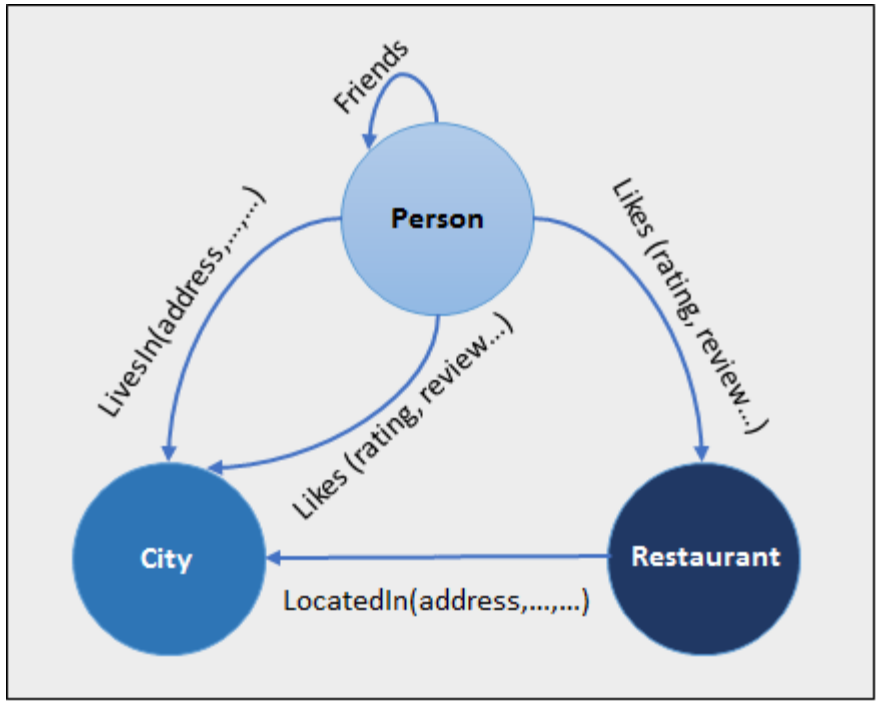
- Graph modeling is an alternative to traditional relational database modeling
- We'll design and implement graphs in both SQL Server and Neo4j, and migrate databases from/to relational sources

After completing this topic:

- you will understand the pro's and con's of graph vs relational data modeling
- you will be able to design, implement and query a graph database using MS SQL Server and Neo4j technologies

Graph vs relational modeling

- Only edges and nodes
- One-to-one
  - Relational: foreign key
  - Graph: edge
- One-to-many
  - Relational: foreign key
  - Graph: edge
- Many-to-many
  - Relational: linking table with combined key
  - Graph: edge



Source: https://docs.microsoft.com/en-us/sql/relational-databases/graphs/sql-graph-sample

Graph extensions on SQL Server

- Node and edge tables with id's generated automatically
- Edges can be attributed and constrained
- Extended SQL syntax for querying
  - Match patterns to follow paths of nodes and edges
- Special graph grouping for the shortest path
- Poor performance on large graphs due to non-native implementation

Neo4j

- An open source Java-based graph database technology
- Data stored in binary node, label, relationship and property files
- A node may have multiple labels (types)
- An edge (relationship) can have only a single type
- Egdes are directed, but direction may be disregarded in queries
- Edge and node properties are key-value pairs
- Native data types are bool, float, integer, point, string, date/time and lists (arrays) of these

The cypher language

- Resembles SQL
- MATCH-WHERE-MAIN_STATEMENT
- The MATCH pattern may contain multiple path specs
- WHERE filters the attributes
- Main statements:
  - Return
  - Delete (nodes, edges)
  - Remove (properties, labels)
  - Set (properties, labels)
  - Create (nodes, edges)
  - Merge (pattern)